Fiscal Year:	FY 2007	**Task Last Updated:**	FY 01/08/2007
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Project Title:	Development of a Gait Adaptability Training Program as a Countermeasure for Postflight Locomotor Dysfunction		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Sensorimotor Adaptation Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77058-3607	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	10/01/2004	**End Date:**	10/01/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/1/2008 because of reporting requirements (3/09 jvp)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Cohen, Helen (Baylor College of Medicine) Mulavara, Ajitkumar (NSBRI)		
Grant/Contract No.:	NCC 9-58-NA00403		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

Following their return to Earth, astronauts experience disturbances in their ability to walk and maintain postural stability due to neural adaptation to the microgravity conditions of space flight. These changes can pose risks to crew safety and mission objectives during planetary EVAs and may impair emergency vehicle egress capability. At present, no operational countermeasure is available to mitigate these risks by facilitating rapid sensorimotor re-adaptation to gravitational environments. Therefore, the goal of this project is to develop both preflight and inflight balance and gait training programs that will facilitate recovery of functional mobility after space flight.
The proposed countermeasure is based on the concept of adaptive generalization. During this type of training the subject gains experience producing the appropriate adaptive motor behavior under a variety of sensory conditions and response constraints. As a result the subject learns to solve a class of motor problems, rather than a specific motor solution to one problem, i.e., the subject learns response generalizability or the ability to "learn to learn." under a variety of environmental constraints. Numerous studies support the concept that motor learning can be enhanced through training but to date practical applications have not been developed.

The Gait Adaptability Training Program will be built upon the existing treadmill exercise program approved for the International Space Station. The Gait Adaptability Program will be done simultaneously with nominal inflight treadmill exercise for long-duration flights on the ISS but can be implemented in a preflight training mode for Lunar missions. By manipulating the sensory conditions of exercise by varying visual flow patterns, body load, and the stability of the walking surface we will systematically and repeatedly challenge the balance control system during treadmill walking. This training regimen will enhance sensorimotor adaptability, facilitating adaptation to gravitational environments.

To develop the Gait Adaptability Training Program we are conducting a series of ground-based studies that will systematically investigate the training efficacy associated with exposing subjects to different combinations of sensory input during treadmill locomotion including alterations in visual flow, body loading, and support surface stability. We will also determine the optimal training mode (strategic vs. adaptive) and schedule (distributed vs. massed) to maximize both the efficacy and efficiency of the training procedure.

In the second year we have had four major accomplishments:

1) We have investigated the efficacy of two different gait adaptability training modes (adaptive vs. strategic). Subjects participating in the strategic training program adapted at a faster rate to a novel sensory environment. This finding has important implications for the development and design of training regimens aimed at improving sensorimotor adaptability in astronauts during gravitational transitions.

2) We have studied the interplay between strategic and adaptive control in gait adaptability training. Results indicate that adaptive recalibration of locomotor function depends on the rate at which strategic control is reduced. The implication of this result from a training perspective is that transition from a strategic to an adaptive control scheme can be characterized and therefore this relationship can be used as a metric to track the efficacy of each training session in a real-time manner and the training modified in a dynamic fashion to match each subject's progress.

3) We have conducted a study examining the use of body unloading as a gait adaptability training modality. The objective of this study was to investigate if limited exposure to unloaded treadmill walking leads to adaptive alteration in locomotor function. Subject's locomotor function was tested before and after 30 minutes of treadmill walking at 40% body weight support produced by a pneumatically controlled harness device integrated into the treadmill system. Data indicate that during post adaptation testing there were adaptive changes in head movement control, lower limb kinematics and gait cycle timing. Therefore, even a limited exposure to body unloading causes adaptive reorganization of multiple locomotor control mechanisms and can play a central role in gait adaptability training programs.

4) We have developed the capability of producing support surface variation during gait training by mounting a treadmill on a six-degree-of-freedom motion device. This hardware development will allow us to evaluate the efficacy of this type of training in conjunction with variation in visual flow and body unloading. Currently, testing is underway to determine the hardware and "human in loop" limitations that will constrain the movement profiles (range of motion, velocity and acceleration) used for our gait adaptability training.

This study is directly related to our current International Space Station study "Mobility" whose goal is to develop an inflight treadmill training program to increase adaptability of locomotor function after space flight. The information from the current NSBRI study will enable us to optimize the inflight training program and schedule. The results from this study will also be used to specify the requirements for future in-flight treadmills used during long-duration space flight.

In the coming year we will continue work on support surface variation as a training modality using the six degree of freedom treadmill locomotor training system. We will be evaluating the efficacy of this type of training in conjunction with variation in body loading and visual flow to increase gait adaptability.

Rationale for HRP Directed Research:

From a clinical perspective exposure to varied sensorimotor stimuli during treadmill walking provides environmental enrichment to enhance gait and postural adaptability. A vast literature demonstrates that the central nervous system requires a rich and varied sensorimotor environment to maintain normal structure and function. Also, a converging body of evidence supports the use of environmental enrichment both before and after brain trauma, as an intervention to help restore behavioral function These studies indicate that exposure to environmental enrichment can play a central role in the restoration of functional mobility. Importantly, studies have shown that training comprised of challenging balance tasks compared to simple repetitive or treadmill exercises alone increase structural plasticity in the motor cortex and thalamus.
Within the context of locomotor functional training and rehabilitation, it is clear that an enriched and challenging environment provided by virtual reality devices provides the optimal conditions to explore variability and in that exploration increase adaptability of motor function. Indeed, the concept of employing virtual reality technology as a tool for rehabilitation has become a topic of great interest among a number of researchers. A collateral benefit of the application virtual reality technology, in this context, will be to make gait training programs for both space flight and rehabilitation more interesting, ultimately leading to increased adherence to prescribed training regimens.

Research Impact/Earth Benefits:

In addition, the gait adaptability training program being developed by this project can be used to reduce the propensity of falling in the elderly. We are currently pursuing this line of research in collaboration with Dr. Ronita Cromwell and

	her graduate student Regina Buccello at the University of Texas Medical Branch in Galveston. This associated study will compare a group of elders receiving the specialized treadmill adaptability training with a group that receives the treadmill training only. Preliminary results are showing significantly improved gait function in elders that receive the training. Since we are developing a training program to improve functional mobility after long-duration space flight the primary users of this research include the astronauts, flight surgeons and trainers. We will use the information obtained in this research to motivate exercise devices and training programs in support of long-duration space flight. This research will also benefit the rehabilitation community by providing new information that can be used in the design of novel sensorimotor variable practice training programs. Additionally, we anticipate that our research will aid in the development of exercise programs that will reduce the propensity to fall in the elderly. Overall, this research serves the general scientific community by providing further insight into the mechanisms underlying adaptive plasticity in the central nervous system.
Task Progress:	1) Completed study investigating relative efficacy of strategic and adaptive gait adaptability training in support of Specific Aim 1. 2) Completed study investigating the interplay between strategic and adaptive control mechanisms in plastic recalibration of locomotor function and have submitted a manuscript for publication in Experimental Brain Research. It has been accepted with minor revisions. 3) Completed study examining if unloaded treadmill walking can be used as a gait adaptability training modality in support of Specific Aim 2. 4) Completed the development of a facility within the NASA-JSC Neuroscience Laboratories, compliant with safety requirements, that will be capable of providing six degree of freedom support surface variation to a subject training on a treadmill as a gait adaptability training modality in support of Specific Aim 2. 5) Completed development of software capable of generating independent six degree of freedom of the treadmill-motion base trainer. This will further the development of support surface motion profiles for use in treadmill training for investigations supporting Specific Aim 2.
Bibliography Type:	Description: (Last Updated: 11/14/2025)
Articles in Peer-reviewed Journals	Mulavara AP, Houser J, Miller C, Bloomberg JJ. "Full-body gaze control mechanisms elicited during locomotion: effects of VOR adaptation." J Vestib Res. 2005;15(5-6):279-89. PMID: 16614474 , Oct-2005
Articles in Peer-reviewed Journals	Mulavara AP, Richards JT, Ruttley T, Marshburn A, Nomura Y, Bloomberg JJ. "Exposure to a rotating virtual environment during treadmill locomotion causes adaptation in heading direction." Exp Brain Res. 2005 Oct;166(2):210-9. PMID: 16034569 , Oct-2005
Articles in Peer-reviewed Journals	Nomura Y, Mulavara AP, Richards JT, Brady R, Bloomberg JJ. "Optic flow dominates visual scene polarity in causing adaptive modification of locomotor trajectory." Brain Res Cogn Brain Res. 2005 Dec;25(3):624-31. PMID: 16216478 , Dec-2005
Articles in Peer-reviewed Journals	Richards JT, Mulavara AP, Bloomberg JJ. "The interplay between strategic and adaptive control mechanisms in plastic recalibration of locomotor function." Experimental Brain Research. Submitted for Publication. September 2006. , Sep-2006
Awards	Bloomberg J. "NASA Performance Award 2006. January 2006." Jan-2006